

IV Curve Diagnosis



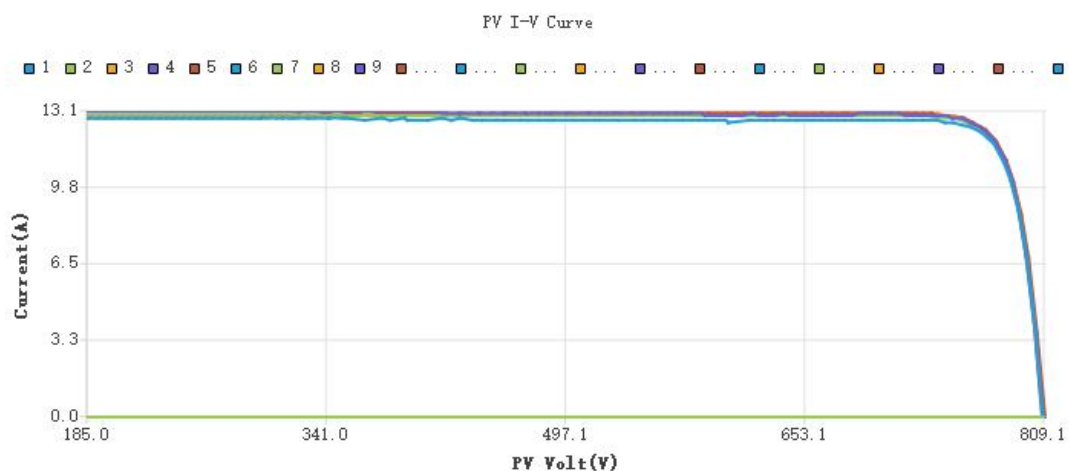
Introduction

In an entire solar system, there are several parts that matters. PV panels are a critical and integral part of the system and their performance can have a direct impact on the output efficiency of the solar system. For example, if some dust block the sunlight reaching the PV panels, the current input to the inverter will be reduced. Therefore, for the conversion efficiency of the system, we need to check the status of the PV modules frequently to diagnose any issue in the very early stage.

However, the photovoltaic panels are installed in open areas and are easily affected by the surrounding environment. Factors such as dust, leaves and other pollutants have the greatest impact on the system's power output. It would be time consuming and inefficient to require personnel to check all the photovoltaic panels each time.

Thus, the IV curve diagnostic function is developed and now applying on our on-grid inverter X1-MINI G4, X1-BOOST G4, X3-MEGA G2 and X3-FORTH.

I-V curve

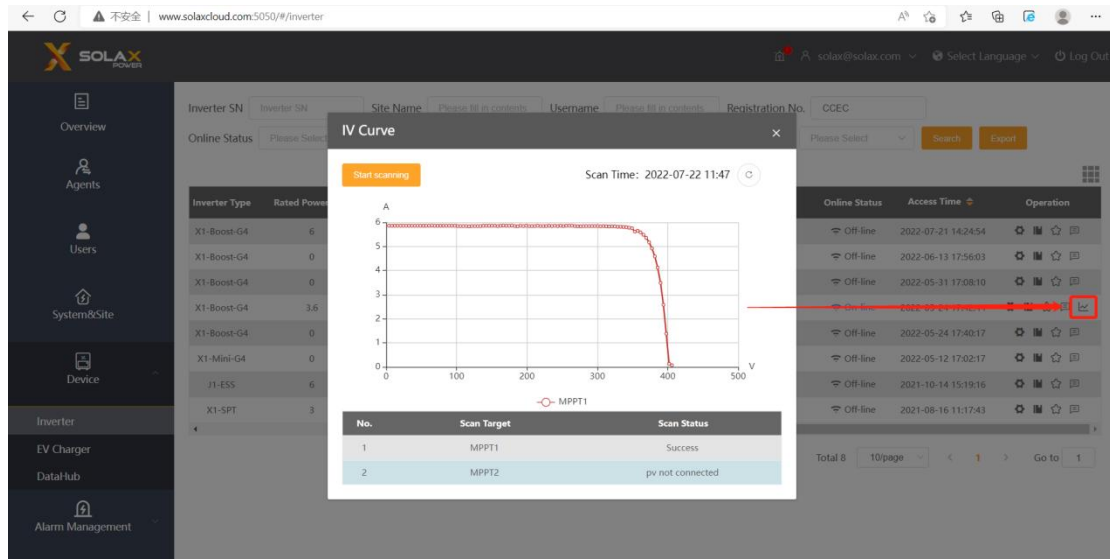


The IV curve has voltage on the horizontal axis and current on the vertical axis, the maximum value on the horizontal axis is the open circuit voltage and the maximum value on the vertical axis is the short circuit current.

The curve under normal conditions can be roughly divided into three sections, the first being nearly horizontal, the second a transitional arc, and the third generally vertical, stretching downwards.

Application

The I-V curve diagnosis function now can be achieved on our platform 'SolaXcloud' or via uplink software.



You can complete the IV curve scan by following the steps below:

- Search for your inverter on the website by its SN number
- Find the icon for the curve scan in the corresponding operation bar (The red boxed position in the picture)
- Click on the icon to go to the IV curve page
- Click on Start Scanning and wait
- The scan duration for each MPPT is 30~60s, then you can see the finished scan curve
- You can view the results of the last 6 scans and export the scanned data (under development)

Notes

- The scan duration for each MPPT is 30~60s, during which the normal operation of the inverter may be affected and the output power may fluctuate a little.
- Ensure the weather is clear and free of cloud cover at the time of scanning.
- The input power per MPPT should be within 16kw when using the IV curve scan function.
- The open circuit voltage should be within 800V when using the IV curve scan function.
- When using the IV curve scan function, the inverter should not be under the derating situation.